Multiple Testing

Pietro Franceschi | pietro.franceschi@fmach.it

Where we left off

In a single test, we accept a false positive rate of $\alpha = 0.05$.

Where we left off

In a single test, we accept a false positive rate of $\alpha = 0.05$.

This means: if H0 is true, we wrongly reject it **5% of the time**.

Where we left off

In a single test, we accept a false positive rate of $\alpha = 0.05$.

This means: if H0 is true, we wrongly reject it **5% of the time**.

The metabolomics reality

We do not test *one* hypothesis. We simultaneously test **hundreds or thousands** of variables.

A concrete example

Imagine a metabolomics experiment:

- **2 groups** of samples (treated vs control)

A concrete example

Imagine a metabolomics experiment:

- **2 groups** of samples (treated vs control)
- **1000 metabolites** measured for each sample

A concrete example

Imagine a metabolomics experiment:

- **2 groups** of samples (treated vs control)
- **1000 metabolites** measured for each sample
- A **t-test** for every metabolite $\rightarrow$ 1000 p-values

A concrete example

Imagine a metabolomics experiment:

- **2 groups** of samples (treated vs control)
- **1000 metabolites** measured for each sample
- A **t-test** for every metabolite $\rightarrow$ 1000 p-values
- We flag anything with $p < 0.05$ as "significant"

A concrete example

Imagine a metabolomics experiment:

- **2 groups** of samples (treated vs control)
- **1000 metabolites** measured for each sample
- A **t-test** for every metabolite $\rightarrow$ 1000 p-values
- We flag anything with $p < 0.05$ as "significant"

A concrete example

Imagine a metabolomics experiment:

- **2 groups** of samples (treated vs control)
- **1000 metabolites** measured for each sample
- A **t-test** for every metabolite $\rightarrow$ 1000 p-values
- We flag anything with $p < 0.05$ as "significant"

How many false positives should we expect?

Expected false positives

If none of the 1000 metabolites truly differ between groups:

$$\text{Expected false positives} = 1000 \times 0.05 = \mathbf{50}$$

Expected false positives

If none of the 1000 metabolites truly differ between groups:

$$\text{Expected false positives} = 1000 \times 0.05 = \mathbf{50}$$

We would report **50 "significant" metabolites** — all of them artefacts.

Expected false positives

If none of the 1000 metabolites truly differ between groups:

$$\text{Expected false positives} = 1000 \times 0.05 = \mathbf{50}$$

We would report **50 "significant" metabolites** — all of them artefacts.

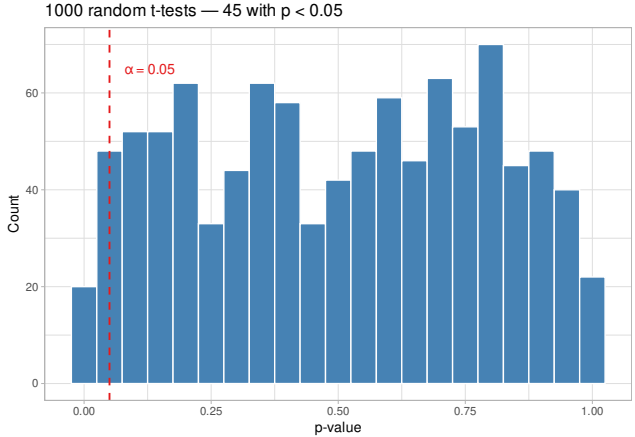
Performing many tests inflates the probability of finding something by chance alone.

Section 1

Visualising the problem

p-values under H_0

When there is **no real effect**, p-values follow a uniform distribution between 0 and 1.



What this tells us

- The histogram is **flat** — every p-value is equally likely under H0

What this tells us

- The histogram is **flat** — every p-value is equally likely under H0
- The red bar on the left ($p < 0.05$) contains ~50 tests

What this tells us

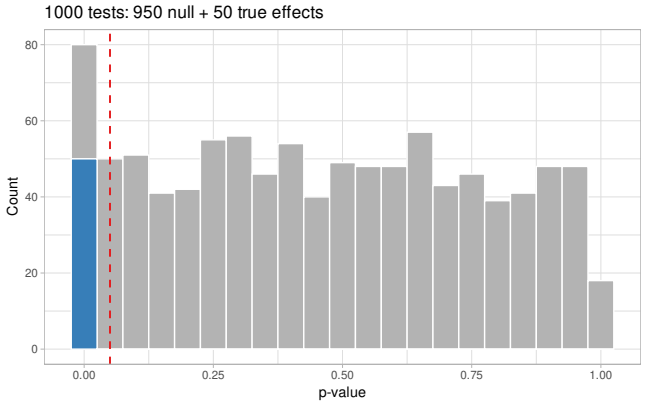
- The histogram is **flat** — every p-value is equally likely under H0
- The red bar on the left ($p < 0.05$) contains ~50 tests
- These are all false positives — there is no true signal in this data

What this tells us

- The histogram is **flat** — every p-value is equally likely under H0
- The red bar on the left ($p < 0.05$) contains ~50 tests
- These are all false positives — there is no true signal in this data
- A flat p-value histogram is actually a useful **quality check**: it confirms no real effects are present

When there ARE real effects

If some variables truly differ between groups, the histogram is no longer flat: a spike appears near zero on top of the uniform background.



Reading the mixed histogram

- The **flat part** across the full range $\rightarrow$ null features (false positives hiding here)

Reading the mixed histogram

- The **flat part** across the full range $\rightarrow$ null features (false positives hiding here)
- The **spike near zero** $\rightarrow$ true positives piling up at small p-values

Reading the mixed histogram

- The **flat part** across the full range $\rightarrow$ null features (false positives hiding here)
- The **spike near zero** $\rightarrow$ true positives piling up at small p-values
- The height of the spike above the flat baseline is proportional to the number of real effects

Reading the mixed histogram

- The **flat part** across the full range $\rightarrow$ null features (false positives hiding here)
- The **spike near zero** $\rightarrow$ true positives piling up at small p-values
- The height of the spike above the flat baseline is proportional to the number of real effects
- A spike + flat pattern is the signature of a real experiment with **some** true signal

Reading the mixed histogram

- The **flat part** across the full range $\rightarrow$ null features (false positives hiding here)
- The **spike near zero** $\rightarrow$ true positives piling up at small p-values
- The height of the spike above the flat baseline is proportional to the number of real effects
- A spike + flat pattern is the signature of a real experiment with **some** true signal

Reading the mixed histogram

- The **flat part** across the full range → null features (false positives hiding here)
- The **spike near zero** → true positives piling up at small p-values
- The height of the spike above the flat baseline is proportional to the number of real effects
- A spike + flat pattern is the signature of a real experiment with **some** true signal

Practical tip

always plot the p-value histogram before applying any correction — it tells you immediately whether your data contains any signal at all.

Section 2

Correction methods

Two families of solutions

1. **Control FWER** — ensure the probability of *any* false positive stays below α

Two families of solutions

1. **Control FWER** — ensure the probability of *any* false positive stays below α
   - Bonferroni, Holm

Two families of solutions

1. **Control FWER** — ensure the probability of *any* false positive stays below α
   - Bonferroni, Holm
   - Very conservative; appropriate when every false positive is dangerous

Two families of solutions

1. **Control FWER** — ensure the probability of *any* false positive stays below α
   - Bonferroni, Holm
   - Very conservative; appropriate when every false positive is dangerous
2. **Control FDR** — ensure the *proportion* of false positives among findings stays below α

Two families of solutions

1. **Control FWER** — ensure the probability of *any* false positive stays below α
   - Bonferroni, Holm
   - Very conservative; appropriate when every false positive is dangerous
2. **Control FDR** — ensure the *proportion* of false positives among findings stays below α
   - Benjamini-Hochberg

Two families of solutions

1. **Control FWER** — ensure the probability of *any* false positive stays below α
   - Bonferroni, Holm
   - Very conservative; appropriate when every false positive is dangerous
2. **Control FDR** — ensure the *proportion* of false positives among findings stays below α
   - Benjamini-Hochberg
   - Less conservative; appropriate for discovery science

Family-wise error rate (FWER)

The probability of making **at least one** false positive across m independent tests:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

Family-wise error rate (FWER)

The probability of making **at least one** false positive across m independent tests:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

Tests (m)	FWER at $\alpha = 0.05$
1	0.05
10	0.40
50	0.92
100	0.994
1000	≈ 1

Bonferroni correction

The simplest FWER control: divide the threshold by the number of tests.

$$\alpha_{\text{Bonf}} = \frac{\alpha}{m}$$

Bonferroni correction

The simplest FWER control: divide the threshold by the number of tests.

$$\alpha_{\text{Bonf}} = \frac{\alpha}{m}$$

For $m = 1000$ and $\alpha = 0.05$:

$$\alpha_{\text{Bonf}} = \frac{0.05}{1000} = 0.00005$$

Bonferroni correction

The simplest FWER control: divide the threshold by the number of tests.

$$\alpha_{\text{Bonf}} = \frac{\alpha}{m}$$

For $m = 1000$ and $\alpha = 0.05$:

$$\alpha_{\text{Bonf}} = \frac{0.05}{1000} = 0.00005$$

Very conservative.

A true positive must produce an extremely small p-value to survive. In metabolomics with correlated variables this is often too strict.

False Discovery Rate (FDR)

Introduced by **Benjamini & Hochberg (1995)** — one of the most cited papers in statistics.

False Discovery Rate (FDR)

Introduced by **Benjamini & Hochberg (1995)** — one of the most cited papers in statistics.

Instead of controlling the probability of *any* false positive, control the **expected proportion** of false positives among all discoveries:

$$\text{FDR} = E\left[\frac{\text{false positives}}{\text{total discoveries}}\right]$$

False Discovery Rate (FDR)

Introduced by **Benjamini & Hochberg (1995)** — one of the most cited papers in statistics.

Instead of controlling the probability of *any* false positive, control the **expected proportion** of false positives among all discoveries:

$$\text{FDR} = E\left[\frac{\text{false positives}}{\text{total discoveries}}\right]$$

The resulting corrected p-values are called q-values.

A q-value of 0.05 means: among all features with $q \leq 0.05$, we expect at most **5% to be false positives**.

Section 3

Summary

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold
- **Bonferroni**: strict FWER control (clinical, confirmatory)

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold
- **Bonferroni**: strict FWER control (clinical, confirmatory)
- **Benjamini-Hochberg**: discovery (metabolomics, exploratory)

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold
- **Bonferroni**: strict FWER control (clinical, confirmatory)
- **Benjamini-Hochberg**: discovery (metabolomics, exploratory)
- Inspect the **p-value histogram**: spike near 0, flat elsewhere

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold
- **Bonferroni**: strict FWER control (clinical, confirmatory)
- **Benjamini-Hochberg**: discovery (metabolomics, exploratory)
- Inspect the **p-value histogram**: spike near 0, flat elsewhere
- Significance is not enough: pair it with an effect size (Cohen's d)

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold
- **Bonferroni**: strict FWER control (clinical, confirmatory)
- **Benjamini-Hochberg**: discovery (metabolomics, exploratory)
- Inspect the **p-value histogram**: spike near 0, flat elsewhere
- Significance is not enough: pair it with an effect size (Cohen's d)

Take-home messages

Warning

never report **uncorrected p-values** when testing many hypotheses

- The **number of tests** matters as much as the threshold
- **Bonferroni**: strict FWER control (clinical, confirmatory)
- **Benjamini-Hochberg**: discovery (metabolomics, exploratory)
- Inspect the **p-value histogram**: spike near 0, flat elsewhere
- Significance is not enough: pair it with an effect size (Cohen's d)

Hundreds of features without correction: a results list dominated by noise. With FDR control, discoveries are principled.